

Pulse Shaping & The Nyquist Criterion

Theoretical Foundations, Implementation, & Laboratory Characterisation

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September 5, 2026

Abstract

In band-limited digital communication systems, transmitting rectangular pulses over physical channels results in temporal pulse spreading, causing Intersymbol Interference (ISI). This report details the theoretical framework of pulse shaping and Harry Nyquist's criterion for zero-ISI transmission. We examine the mathematical formulation of the ideal minimum-bandwidth sinc pulse, analyze its physical limitations, and transition to the practical Raised Cosine (RC) and Root Raised Cosine (RRC) filters. The report provides a rigorous analytical comparison of their spectral properties, details the cascading matched-filter setup designed to maximize receiver Signal-to-Noise Ratio (SNR), and outlines evaluation methodologies including eye diagrams, delay compensation, and timing sensitivity.

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1 Intersymbol Interference (ISI) and Channel Constraints

In digital baseband and passband communication systems, binary data is mapped onto physical wave pulses before transmission over a medium. A naive approach uses rectangular pulses to represent symbols. However, physical communication channels (such as wireless bands, coaxial lines, or optical fibres) exhibit finite physical bandwidth, acting as low-pass filters.

When a rectangular pulse with sharp transitions is passed through a band-limited channel, its high-frequency components are cut off, causing the pulse to smooth out and disperse in the time domain. As a result, the tail of a transmitted pulse spreads into adjacent symbol slots. When consecutive pulses are transmitted, their tails overlap at the receiver, as illustrated below:

$$y(t) = \sum_{k=-\infty}^{\infty} I_k h(t - kT_s) + n(t) \quad (1)$$

where I_k represents the k -th transmitted symbol, T_s is the symbol period, $h(t)$ is the composite system impulse response, and $n(t)$ is additive noise. Sampling the received signal at an arbitrary instant $t = mT_s$ yields:

$$y(mT_s) = I_m h(0) + \sum_{k \neq m} I_k h((m - k)T_s) + n(mT_s) \quad (2)$$

The first term, $I_m h(0)$, is the desired current symbol. The second term, $\sum_{k \neq m} I_k h((m - k)T_s)$, represents the cumulative interference from all other transmitted symbols. This distortion is termed **Intersymbol Interference (ISI)**. If left uncontrolled, ISI closes the margin of error at the receiver, dramatically increasing the Bit Error Rate (BER) even in noise-free channels.

2 The Nyquist Criterion for Zero ISI

To eliminate ISI, Harry Nyquist established that pulses do not need to be strictly confined within their individual symbol slots. Instead, we only require that their cumulative effect at the **exact sampling instants** is zero.

2.1 Time-Domain Formulation

A system achieves zero ISI if its overall impulse response $h(t)$ satisfies:

$$h(nT_s) = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases} \quad (3)$$

This means that at $t = mT_s$, the impulse responses of all adjacent symbols ($k \neq m$) cross exactly through zero, leaving the current symbol entirely free of interference.

2.2 Frequency-Domain Formulation

Applying the Fourier transform to the time-domain zero-ISI condition reveals the frequency-domain Nyquist pulse-shaping criterion. The periodic summation of the system's overall frequency response $H(\omega)$ over frequency shifts of $2\pi/T_s$ must yield a flat, constant response:

$$\sum_{k=-\infty}^{\infty} H\left(\omega + \frac{2\pi k}{T_s}\right) = T_s \quad \text{for } |\omega| \leq \frac{\pi}{T_s} \quad (4)$$

Any filter $H(\omega)$ satisfying this equation will naturally possess zero-crossings at every non-zero integer multiple of the symbol period T_s in the time domain.

3 The Sinc Pulse: Minimum Bandwidth Solution

The most mathematically direct way to satisfy the frequency-domain flat-summation requirement is to design $H(\omega)$ as an ideal brick-wall low-pass filter (LPF) that truncates sharply at the Nyquist frequency $W = \pi/T_s$ (or $f_N = 1/(2T_s)$ Hz):

$$H(\omega) = \begin{cases} T_s & |\omega| \leq \frac{\pi}{T_s} \\ 0 & |\omega| > \frac{\pi}{T_s} \end{cases} \quad (5)$$

The inverse Fourier transform of this ideal filter yields the famous **sinc pulse** in the time domain:

$$h(t) = \text{sinc}\left(\frac{t}{T_s}\right) = \frac{\sin(\pi t/T_s)}{\pi t/T_s} \quad (6)$$

This pulse achieves zero ISI with the absolute minimum physical bandwidth required for a given transmission rate, known as the **Nyquist Bandwidth**:

$$B_{\min} = \frac{1}{2T_s} \text{ Hz} \quad (7)$$

3.1 Practical Limitations of Sinc Pulse Shaping

Although spectrally optimal, the ideal sinc pulse is unusable in practical systems for two primary reasons:

1. **Non-Causality (Unrealizability):** The impulse response $h(t)$ starts at $t = -\infty$ and ends at $t = +\infty$. A causal, physically realizable system must have $h(t) = 0$ for $t < 0$. Truncating the sinc pulse to make it causal introduces sharp spectral transitions, violating the zero-ISI condition and causing massive out-of-band energy leakage.
2. **Extreme Sensitivity to Timing Jitter:** The tails of the sinc pulse decay very slowly at a rate proportional to $1/t$. If the receiver's sampler suffers from even a minor clock synchronization error or timing jitter, it will sample slightly off the ideal zero-crossings. Due to the slow $1/t$ decay, the offsets of thousands of adjacent symbols can accumulate, resulting in an infinite sum of ISI that completely closes the decision window.

4 Practical Raised Cosine (RC) Pulse Shaping

To overcome the physical limitations of the sinc pulse, practical systems use a filter where the frequency response rolls off gradually to zero instead of cutting off abruptly. The most popular filter family for this is the **Raised Cosine (RC) filter**.

4.1 Mathematical Equations

The frequency response $H_{RC}(\omega)$ of a Raised Cosine filter is defined as:

$$H_{RC}(\omega) = \begin{cases} T_s & 0 \leq |\omega| \leq (1 - \alpha)W \\ \frac{T_s}{2} \left\{ 1 - \sin \left[\frac{\pi}{2\alpha W} (|\omega| - W) \right] \right\} & (1 - \alpha)W \leq |\omega| \leq (1 + \alpha)W \\ 0 & |\omega| > (1 + \alpha)W \end{cases} \quad (8)$$

where $W = \frac{\pi}{T_s}$ and α is the **roll-off factor** ($0 \leq \alpha \leq 1$).

The corresponding time-domain impulse response is:

$$h_{RC}(t) = \text{sinc} \left(\frac{t}{T_s} \right) \left[\frac{\cos \left(\frac{\pi \alpha t}{T_s} \right)}{1 - \left(\frac{2\alpha t}{T_s} \right)^2} \right] \quad (9)$$

4.2 The Roll-off Factor (α) and Trade-offs

The roll-off factor (α), sometimes denoted as β , represents the fractional excess bandwidth used by the filter beyond the theoretical Nyquist minimum:

$$B_{\text{actual}} = \frac{1 + \alpha}{2T_s} \text{ Hz} \quad (10)$$

- **When $\alpha = 0$:** The filter becomes the ideal minimum-bandwidth sinc pulse, requiring $B = \frac{1}{2T_s}$ Hz, but exhibiting slow $1/t$ decay and non-causality.
- **When $\alpha = 1.0$:** The filter is called the full-cosine roll-off filter. It requires double the minimum bandwidth ($B = \frac{1}{T_s}$ Hz), but its time-domain impulse response decays incredibly fast at a rate proportional to $1/t^3$.

The fast $1/t^3$ decay rate dramatically improves tolerance to receiver timing jitter. If a minor sampling offset occurs, the fast-decaying tails of adjacent symbols do not sum to a destructive value, keeping the ISI penalty negligible.

4.3 Numerical Singularity Handling in Implementation

When writing simulation code, implementing the analytical formula of the Raised Cosine pulse directly leads to division-by-zero errors. There is a **removable singularity** in the denominator of the decay multiplier at $1 - (2\alpha t/T_s)^2 = 0 \implies t = \pm \frac{T_s}{2\alpha}$.

To make the code numerically stable, a conditional statement must analytically handle these limits:

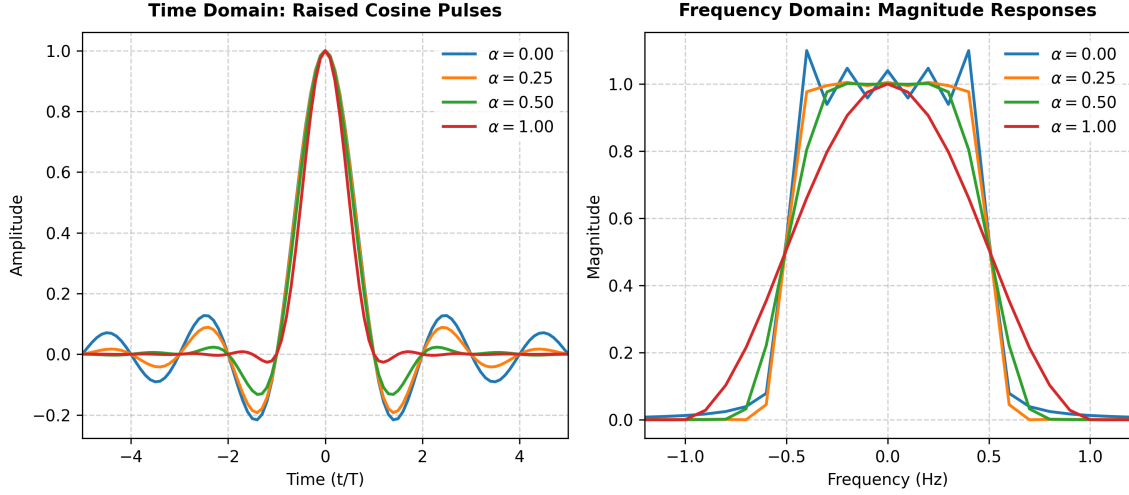


Figure 1: Raised Cosine Pulse Response in Time Domain (left) and Frequency Domain (right) for varied roll-off factors $\alpha \in \{0, 0.25, 0.5, 1.0\}$. Note how a larger α smooths the spectral transitions and causes the time-domain tails to decay much faster.

$$h_{RC}\left(\pm\frac{T_s}{2\alpha}\right) = \frac{\pi}{4}\text{sinc}\left(\frac{1}{2\alpha}\right) \quad (11)$$

5 Root Raised Cosine (RRC) and Matched Filter Cascading

In real communication links, the physical channel introduces additive noise. To maximize the Signal-to-Noise Ratio (SNR) at the receiver and protect the digital stream from Additive White Gaussian Noise (AWGN), we must apply **matched filtering**. Matched filter theory states that the receiver filter's frequency response must be the complex conjugate of the transmitter's pulse shape:

$$H_{Rx}(\omega) = H_{Tx}^*(\omega) \quad (12)$$

To achieve both ****maximum SNR (matched filtering)**** and ****zero ISI (raised-cosine overall system response)****, we split the overall Raised Cosine response equally between the transmitter and the receiver:

$$H_{RC}(\omega) = H_{Tx}(\omega) \cdot H_{Rx}(\omega) \implies H_{Tx}(\omega) = H_{Rx}(\omega) = \sqrt{H_{RC}(\omega)} \quad (13)$$

This square-root filter is called the **Root Raised-Cosine (RRC)** filter.

5.1 RRC Filter Formulations

The frequency response of an RRC filter is $H_{RRC}(\omega) = \sqrt{H_{RC}(\omega)}$. Its time-domain impulse response is:

$$h_{RRC}(t) = \frac{4\alpha}{\pi\sqrt{T_s}} \frac{\cos\left(\frac{(1+\alpha)\pi t}{T_s}\right) + \frac{T_s}{4\alpha t} \sin\left(\frac{(1-\alpha)\pi t}{T_s}\right)}{1 - \left(\frac{4\alpha t}{T_s}\right)^2} \quad (14)$$

Because we split the filter in the frequency domain, the RRC pulse itself **does not satisfy the zero-ISI condition** on its own. Its time-domain response has non-zero crossings at symbol intervals. Zero ISI is only restored after the signal passes through both the Tx and Rx RRC filters in cascade. The convolution of the two RRC filters yields the Raised Cosine response:

$$h_{cascaded}(t) = h_{RRC}(t) * h_{RRC}(t) = h_{RC}(t) \quad (15)$$

This convolution property is illustrated in Figure 2.

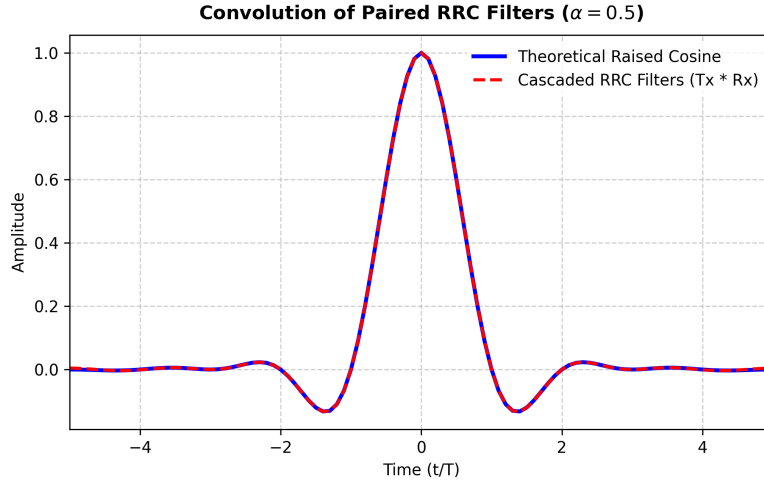


Figure 2: Convolution of two Root Raised Cosine (RRC) filters with $\alpha = 0.5$. The cascaded response (red dashed) perfectly matches the analytical Raised Cosine pulse (blue), confirming that zero-ISI properties are restored after matched filtering.

5.2 RRC Removable Singularities

Similar to the RC filter, the analytical equation for the RRC pulse contains removable singularities that cause division-by-zero errors in computer simulations:

- At $t = 0$:

$$h_{RRC}(0) = 1 - \alpha + \frac{4\alpha}{\pi} \quad (16)$$

- At $t = \pm \frac{T_s}{4\alpha}$:

$$h_{RRC}\left(\pm \frac{T_s}{4\alpha}\right) = \frac{\alpha}{\sqrt{2}} \left[\left(1 + \frac{2}{\pi}\right) \sin\left(\frac{\pi}{4\alpha}\right) + \left(1 - \frac{2}{\pi}\right) \cos\left(\frac{\pi}{4\alpha}\right) \right] \quad (17)$$

A robust software implementation must check for these singularities and handle them conditionally.

6 System Performance Evaluation Tools

To evaluate how well a pulse-shaping system performs under noise, timing offsets, and channel distortions, several graphical and analytical tools are used.

6.1 The Eye Diagram

The eye diagram is the primary visual tool for evaluating digital baseband quality. It is generated by slicing the continuous filtered received signal into overlapping segments of $2T_s$ duration and displaying them together on a plot.

- **Vertical Eye Opening (Eye Height):** Represents the margin of safety against additive noise. A wide-open eye indicates low noise and low ISI. The exact center of the eye corresponds to the optimal sampling instant where SNR is maximized.
- **Horizontal Eye Width:** Represents the margin of safety against timing jitter. A narrow horizontal opening means minor timing offsets will quickly induce bit errors.
- **Effect of Distortions:**
 - **AWGN:** Blurs the traces vertically and horizontally, narrowing the eye.
 - **Multipath Channels:** Multipath transmission occurs when a signal arrives at the receiver via multiple paths of different delays (e.g., a primary path plus a delayed echo, modeled as $[1.0, 0.4]$). These delayed echoes ruin the carefully designed zero-crossing properties of the RRC cascade, causing the eye diagram to close vertically and horizontally due to severe, irreducible ISI.

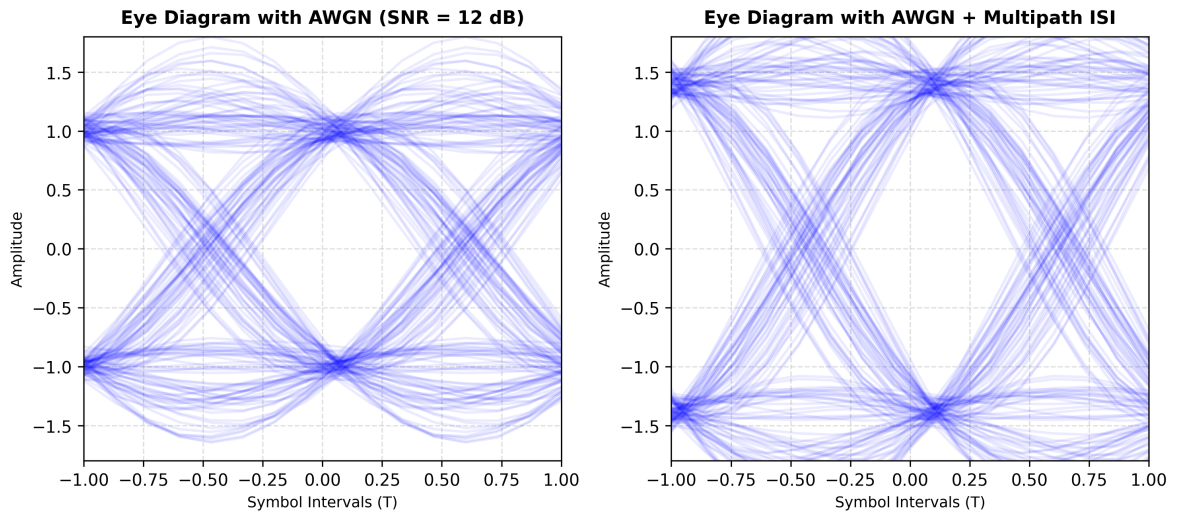


Figure 3: Comparison of simulated eye diagrams for BPSK signaling. Left: System with AWGN (SNR = 12 dB) showing a clean, wide-open eye with zero ISI. Right: System with AWGN and Multipath ISI (channel coefficients $[1.0, 0.4]$) showing severe eye closure and signal smearing.

6.2 Delay Compensation in Simulations

Finite Impulse Response (FIR) digital filters introduce a physical propagation delay. For an FIR filter with a span of S symbols and an oversampling factor of SPS (samples per symbol),

the peak response is delayed by exactly half of the filter span:

$$\text{Delay}_{\text{Filter}} = \frac{\text{span} \times \text{SPS}}{2} \text{ samples} \quad (18)$$

When cascading a transmitter RRC filter and a receiver RRC filter, their delays add up, resulting in a total cascade delay of:

$$\text{Delay}_{\text{Total}} = \text{span} \times \text{SPS} \text{ samples} \quad (19)$$

In computer simulation, the receiver must discard the first $\text{Delay}_{\text{Total}}$ transient samples before slicing the signal to make decisions. Failing to compensate for this delay causes the receiver to sample the transient window, resulting in a random decision state and a catastrophic 50% Bit Error Rate.

6.3 Bit Error Rate (BER) Sensitivity to Timing Offsets

Sampling the filtered waveform at the exact center of the eye yields the optimal BER. If the receiver's clock is offset by a fraction of a sample, the sampling instant moves away from the zero-crossings. The receiver is exposed to the rising tails of adjacent symbols, which rapidly closes the margin of safety. Plotting the BER on a logarithmic scale against timing offsets reveals a sharp, V-shaped curve, proving that minor synchronization errors will destroy system performance.

7 Conclusion & Summary of Pulse Shaping Design Parameters

The design of a communication link's pulse shaping filter represents a careful balancing act between bandwidth efficiency, hardware complexity, and robustness to synchronization errors. The core system behaviors are summarized below:

Pulse Shape	Bandwidth Required	Tail Decay Rate	Jitter Tolerance	Zero-ISI Condition
Ideal Sinc Pulse	Minimum: $1/(2T_s)$	Slow: $1/t$	Extremely Poor	Satisfied on its own
RC ($\alpha = 0.25$)	Low: $1.25/(2T_s)$	Moderate: $1/t^3$	Fair	Satisfied on its own
RC ($\alpha = 0.50$)	Moderate: $1.50/(2T_s)$	Fast: $1/t^3$	Good	Satisfied on its own
RC ($\alpha = 1.00$)	Maximum: $1/T_s$	Very Fast: $1/t^3$	Excellent	Satisfied on its own
RRC (Any α)	Split: $\sqrt{H_{RC}(\omega)}$	Complex	Poor (Uncascaded)	Only after cascading

Table 1: Properties of different pulse-shaping filters and roll-off factors.

By splitting the Raised Cosine response into Root Raised Cosine filters at the transmitter and receiver, modern digital systems achieve the dual objectives of **distortionless (zero-ISI) transmission** and **optimal noise immunity (matched filtering)**. These principles form the bedrock of high-speed transmission standards such as Wi-Fi, LTE, and 5G.